Phase 1

MATH2319 - Machine Learning

Course Project Phase 1

Assessment Weight: 15% (Marked out of 100 points)

Project Group Membership Finalisation Due: Sunday, March 29, 11:59 pm

Phase 1 Report Due: Sunday, April 19, 11:59 pm (Online submission)

Feedback Mode

Feedback will be provided using Canvas.

Learning Objectives Assessed

This assessment task supports CLOs 1, 2, 3, 4.

Summary

In this course, you will be required to complete a major group course project. For your project, you can work alone if you like; or you can work in groups of 2 or 3 students (and no more). The idea with the course project will be to apply what you have learned in class and more importantly, **to have fun**! The project shall entail the following tasks.

Project Phase 1 (This Assessment) :

Data pre-processing (dealing with missing values, dropping ID-like columns, data aggregation, etc.) as appropriate.

Data exploration and visualisation (charts, graphs, interactions, etc) as appropriate.

Project Phase 2:

Predictive modelling as appropriate.

Purpose

The purpose of the course project is to apply machine learning techniques **to a problem of your own choosing** using Python and the Scikit-Learn module. The project will be hands-on, and it will require you to select appropriate performance metrics for your problem and also properly compare the performance of different methods. You will also gain experience in documenting your findings and your code in a Jupyter Notebook environment. In addition, you will get the opportunity to receive feedback on your performance and the report that you submit.

Group Selection Instructions

1. **For the course project, you can work alone, or you can work in groups of 2 or 3 students (and no more). If your group has at least 2 members, only one member must submit.**
2. **Due to some technical issues with Canvas, even if you are working alone, you must still sign up for a group (which will obviously have only one member). If you don't, your project mark may show up as zero in the system, even if you make a personal project submission due to Canvas software glitches.**
3. In order to sign up for groups, please follow the instructions below:
    1. Go to the "People" link on the left panel on Canvas.
    2. Click on the "**ML Project Groups**" tab. You will see a bunch of groups in there.
    3. **Starting from the top, please pick the first empty group number for your group.**
4. You can sign up for any group yourself without permission from the course staff. You can also move between groups as many times as you want before the due date for finalising group membership.
5. **If you are joining a non-empty group, you must have the consent of the existing group member(s) before you sign up.**
6. The due date for finalising your group members will be the **end of Week 4.** Beyond this date, group membership cannot be changed. This is to avoid drama caused by last-minute group changes.
7. All group members shall receive the same mark. Any serious disagreement among team members should be reported to the Course Coordinator, and such disagreements shall be dealt with on a case-by-case basis.

Course Project Instructions and Assessment Criteria:

Compliance with University Policies

1. You may not submit any previous work from RMIT or any other place for this course project, even if it was done by yourself. RMIT policy is that a particular work can be submitted only once for credit.
2. You may not use the datasets in the inspiration projects, nor any dataset we use in this course. These specific datasets are included in the list of "**Blacklisted Datasets**" below.

Programming Language Requirements

The project must be done in Python 3.6 or above.

Submission Instructions

NOTE 1: Canvas can occasionally become a rather unreliable piece of software, unfortunately. So please follow all technical submission instructions for your assessments so that we can correctly identify your files for marking purposes. Thank you.

NOTE 2: If you have 2 or 3 members in your group, then **ONLY ONE** of the members must submit.

NOTE 3: For both reports, please make sure you **include your group name as well as full name(s) and student ID(s) of all group members** (**EVEN IF** you are the only person in the group). Thank you.

WARNING 1: We cannot run plagiarism check software on zipped submissions, so zipped submissions will NOT be marked, and they will not receive any points.

WARNING 2: We require an HTML file for a plagiarism check. This is because we cannot run plagiarism check software on Jupyter notebook files, so submissions missing the HTML file will NOT be marked, and they will not receive any points.

Suppose you are in Group 23.

For Phase 1: You need submit the following 3 files (please do **not** zip them):

1. **Phase1_Group23.csv** (your Phase 1 input data as a **single** CSV file. The idea here is that we should be able to replicate your Phase 1 results successfully by running your notebook with this input CSV file.)
2. **Phase1_Group23.ipynb** (your Jupyter notebook file)
3. **Phase1_Group23.html** (an HTML file directly compiled from your Jupyter notebook file)

NOTE 4: Canvas File Renaming: If you resubmit a file on Canvas, say a file called "Phase1_Group23.html", Canvas will rename it to "Phase1_Group23-2.html" in the second submission, and then to "Phase1_Group23-3.html" in the third submission, etc. That will be OK as far as file naming is concerned, because this is how Canvas works.

Other technical instructions you need to follow while submitting your course project reports as well as applicable penalties are explained in detail on this page.

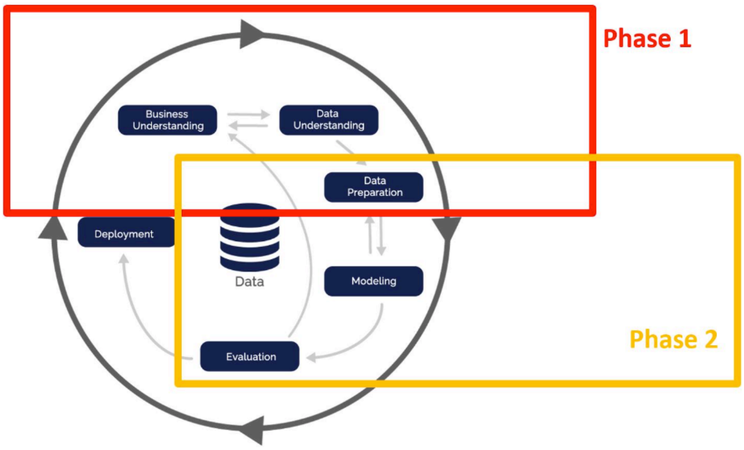
Project Phases

Phase 1 Due at the End of Week 6 (15%):

- Data cleaning and preprocessing (dealing with missing values, dropping ID-like columns, data aggregation, etc.) as appropriate.

- Data exploration and visualisation (charts, graphs, interactions, etc) as appropriate.

This is how CRISP-DM looks in your course project:



Some Terminology

Many names for the same things:

- Some names for **columns in a dataset**: features/ attributes/ variables
- Some names for **rows in a dataset**: instances/ observations/ records
- Some other names for the **target feature** (this is the feature you will predict): response variable/ dependent variable
- Some other names for **descriptive features** (these are the features you will use for making predictions): independent variables/ explanatory variables

Dataset Requirements & Restrictions

1. **Dataset selection:** This is a loosely defined project to give you the maximum level of flexibility. In particular, you will need to choose a project dataset yourself.
2. **Problem type:** Your project needs to be on a **supervised** machine learning problem. That is, you need to have a target feature that needs to be predicted. Both **classification** (categorical target feature) and **regression** (numeric target feature) problems will be accepted.
3. **Dataset resources:** There are no hard restrictions on the dataset that you can select for your project. You can choose a public dataset from popular data repositories, or you can find some other suitable dataset from any website on the Internet or whatever. You can also use data from your work. Please check the bottom of this page for some suggested resources for finding a suitable dataset for your project.
4. **Guidance for selecting a dataset:** As a friendly piece of advice, you might want to select a dataset in line with your future career plans and the particular industries you are interested in. For example, if you plan on working in the banking industry (or conducting academic research in this area), you might want to select a finance-related dataset so that your course project can be a talking point during your interviews.

5. **Blacklisted datasets:** The dataset you choose apparently needs be appropriate for a major machine learning course project. For instance, the following datasets are blacklisted:
   - Boston Housing/ Iris/ Titanic/ US Adult Income Dataset/ Wine/ Wisconsin Breast Cancer
   - Clarification: The "Diamond Prices" dataset is NOT blacklisted - you can use it in your projects.
6. **Minimum requirements: Your dataset must have at least 200 rows and at least 8 descriptive features.** To be clear, your dataset needs to have at least 8 descriptive features **after** dropping all unnecessary features (that you will not use) but **before** one-hot-encoding of any categorical descriptive features. There is no requirement on the data types of your descriptive features: they can be all numeric, all categorical, or a mix.
7. **Random sampling for very large datasets:** There is no upper limit on the number of rows, but if your dataset has more than 5000 rows, you might want to select a random subset with at most 5000 rows so that you do not fry your laptop! That is, you will not lose any points for selecting only a relatively small subset of rows in case your dataset has too many rows.
8. **Exclusion of time series datasets:** In general, the rows in your dataset need to be independent (without any correlations) for ML algorithms to work properly. As an example, if you think about Google's daily stock prices (as in "one day per row"), you will see that these observations are correlated. On the other hand, if you think about customers of a bank (as in "one customer per row"), these observations are independent and uncorrelated (for the most part). **For this reason, datasets for which the rows are correlated such as time series datasets will not be eligible for your course project.** There are ways to tackle this time series dependency within a machine learning setting, but they are beyond the scope of this course.
9. **Competition datasets:** Kaggle or similar competition datasets will be accepted provided that the work you submit is your own personal work.

Some Resources for Data

- **Try this! Google's dataset search tool is here.** You can search for any subject/ keyword you like.
- UCI ML Repository is **here**.
- KEEL Repository is **here**.
- Open ML Repository is **here**.
- GitHub page **here** hosts some datasets.

Project Submission Modes

For your course project, you will have the option to choose from two different submission modes: **Advanced Submission** or **Regular Submission**:

- **Advanced Submission Mode:** The purpose of this mode is to identify truly "high distinction" submissions. Specifically, this mode is for advanced students who are willing to put in extra effort for the full 100 points by undertaking additional tasks. Specifically, you will be required to complete two additional tasks as follows in the Advanced Submission mode:

- **Additional Task for Phase 1 Advanced Mode:** A Literature Review section with at least 600 words (worth 15 points out of 100)
- **Additional Task for Phase 2 Advanced Mode:** An additional section on Artificial Intelligence applied to your particular problem with a narrative of at least 600 words (worth 15 points out of 100)

Details of these two additional tasks are provided in the relevant Assessment Criteria sections below.

- **Regular Submission Mode:** For this mode, you will **not** be required to undertake any one of the above additional tasks.
- **Bonus Points for Regular Submissions:** Occasionally, we will come across regular submission reports that exceed the expectations for this assessment in the regular submission mode. To differentiate these reports and recognise your extra hard work in such cases in the regular submission mode, we will be awarding discretionary bonus points for regular submissions, not to exceed 5 points. The amount of bonus shall be determined by the assessment marker(s), who will try to be as fair as possible across the entire batch of submissions in the class.
- **Clarification 1:** You can make an advanced submission for ***both phases*** or ***just for a single phase***. That is, if you like, you can make a regular submission for Phase 1 and an advanced submission for Phase 2, or vice versa.
- **Clarification 2:** Suppose yours is a **regular** submission. That is, you did **not** undertake the relevant additional task above in your report. In this case, the maximum score you can get will be **90 out of 100** for your submission (up to 85 base points for the report itself plus up to 5 points in bonuses).
- **Clarification 3:** The amount of bonus points will be determined on a case-by-case basis and they can be different for each phase report you submit. For instance, you might get no bonus points for your Phase 1 report, but the full 5 points for your Phase 2 report.
- **Clarification 4:** Your report will be eligible for some bonus points only if your base score (without any bonuses) is the full 85/100. That is, if your report's base score is below 85/10, your report will not be eligible for any bonus points. This is because the bonus points will be reserved only for truly outstanding project reports.
- **Clarification 5:** Suppose yours is an **advanced** submission. That is, you undertook the relevant additional task above in your report. In this case, the maximum score you can get will be **100 out of 100** for your submission.

Project Instructions

1. **Data preprocessing is more than just filling in missing values or removing outliers.** We suggest you have a look at Chapter 3 in the textbook (**"Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies"**, John D. Kelleher, Brian Mac Namee, and Aoife D'Arcy, **First** Edition, MIT Press, 2015) that discusses data preprocessing.

2. Your Phase 1 report needs to contain a certain minimum number of different plots as part of the data exploration and visualisation tasks. Please see the **Plotting Requirements** section below for the specific minimum number of different plots you need to include in your report.
3. In your Phase 2 report, you must try a certain minimum number of different ML algorithms on your problem, perform some sort of feature selection & hyper-parameter tuning, and you must also present performance statistical comparisons to indicate which method seems to work best. If you like, you may also use algorithms not covered in class.
4. **Your narrative (that is, the text that you write, excluding any Python code) in each one of your project reports must be at least 1500 words (plus an additional 600 words for Advanced Submission Modes).** Ideally, it should be less than 3500 words (plus an additional 600 words for Advanced Submission Modes). As a clarification, Python code comments do not count as narrative. This limit is to ensure that your report contains adequate material for a major course project.
5. **To be clear, our expectation here is professional project reports, not a quick and dirty summary of how your Python code works. For more information on this, please see the "Good Writing Practices" Section on this page.**
6. You **are allowed** to use code in our course materials as well as code on our website (www.featureranking.com) with proper acknowledgement.
7. **Your report will be checked against plagiarism using Turnitin software.** Some similarities in computer code will be OK, but excessive copy & paste of text from other people's work (**INCLUDING our course material as well as our website**) will be closely scrutinised. **So, please do not submit our case study back to us as your course project by simply plugging in your dataset!!** This will be very uncool and it will be plagiarism. **That is, your project report must be in your own words.** For additional important information on what constitutes plagiarism, please see the "Integrity and Plagiarism" Section in our Course Policies **here**.
8. Any suspected cases of plagiarism will be dealt with on a case-by-case basis in accordance with RMIT's plagiarism policies together with the penalties that come with these policies. A possible outcome might be that you receive a mark of zero for your submission due to plagiarism. Depending on the nature of the plagiarism, more severe penalties might also apply.
9. Your reports for both phases need to contain a Table of Contents. We are aware that sometimes Table of Contents in Jupyter notebooks are not clickable due to some technical issues. So it would be nice if your Table of Contents were clickable, but this is not a requirement.
10. You need to include your source code in your reports. We cannot mark any project reports as missing the source code. In addition, you will not get any credits for computer code that you do not execute: you will need to show us the output of your code.
11. Submissions are to be made via Canvas by the due date and time.
12. **Please see Course Policies for late report submission penalties.**
13. If you make multiple submissions, only the most recent one shall be marked all the previous submissions will be discarded.

Plotting Requirements for Phase 1

The minimum number of different plots you need to include in your Phase 1 report **for the data exploration & visualisation part** as follows:

- A group of 1 member: **at least 2 plots** of each one of the following: one-variable plots, two-variable plots, and three-variable plots (that is, minimum 2*3=6 plots in total)
- A group of 2 members: **at least 3 plots** of each one of the following: one-variable plots, two-variable plots, and three-variable plots (that is, minimum 3*3=9 plots in total)
- A group of 3 members: **at least 4 plots** of each one of the following: one-variable plots, two-variable plots, and three-variable plots (that is, minimum 4*3=12 plots in total)

Additional plotting instructions are as follows:

1. For plotting, you are free to use whatever Python module you like: Matplotlib, Seaborn, Altair, Plotly, etc.
2. Your plots must be meaningful and they need to make sense with respect to the goals and objectives of your project.
3. As long as your plots are meaningful and relevant, there are no restrictions on the plot types. That is, you can have a mix of box-plots, histograms, line plots, scatter plots etc.
4. For each plot in your report, you will need to label the x- and y-axes as appropriate and also add a meaningful title.

Phase 1 Assessment Criteria (100 points in total)

REQUIREMENT: The green-highlighted texts below must be the Section headers in your Phase 1 report.

- **(Task 0.1) (2 points)** Please give your project a proper title. Example: "Predicting Melbourne House Prices" etc.
- **(Task 0.2) (5 points)** Please include a proper Table of Contents.
- **(Task 1) (28 points total) Introduction:** This section needs to include the following subsections:
  - **(Task 1.1) (3 points) Dataset Source:** Please explain your data source properly and provide a proper citation in a References section at the end of your report.
  - **(Task 1.2) (5 points) Dataset Details:** Explain in sufficient detail what the dataset is about. Also please report the number of observations (rows) and the number of features (columns) in your dataset. **And please print 10 random observations (rows) from your dataset. Please make sure we can see all the columns by adding this code in your report:**
  - ```
    import pandas as pd
    pd.set_option('display.max_columns', None)
    ```
  - **(Task 1.3) (18 points) Dataset Features:** Explain the features in your dataset that you will be including in your project in a table format. To be clear, please include only the features you will use in your project and leave out the ones you will not be using in your project. **In your table, you need to have one feature per row with the following 4 columns:**
    1. Name of the feature
    ```

2. Data type (numeric, binary, nominal categorical, ordinal categorical, date)
3. Units ("Unknown" or "NA" for Not Applicable are acceptable.
4. Brief description

- **(Task 1.4) (2 points) Target Feature:** This is **really** important: you need to clearly identify your target feature. That is, what is it that you will be predicting in your Phase 2 report? Also state if it is a numerical or categorical feature.

- **(Task 2) (7 points) Goals & Objectives** for modelling this particular data.
- **(Task 3) (15 points) Data Cleaning & Preprocessing** as appropriate (dealing with missing values & outliers & incorrect values (such as negative age), dropping ID-like columns, data aggregation if necessary etc). If your dataset is already clean and ready for modelling, you will get the full score for this task, our treat! For this section, you will need to add subsections as appropriate for better organisation of your work.
  - **Task 3 Clarification #1:** Please do **not** perform any one-hot-encoding or scaling in Phase 1. Thank you.
  - **Task 3 Clarification #2:** For this task, **sampling is optional**: you can perform sampling (for a smaller dataset for ease of computation) either in Phase 1 or Phase 2, or you may choose to skip sampling all together if you like.
- **(Task 4) (15 points) Data Exploration & Visualisation** as appropriate with proper explanation: charts, graphs, boxplots, numerical summaries, etc. For this section, you will need to add subsections as appropriate for better organisation.
- **(Task 4.5) (Advanced Submission Mode Only, 15 points) Literature Review** This section needs to report and explain existing closely-related research studies in the particular field of your project. Please make sure you include some recent references published in the last few years. This section needs to include a **minimum of 10 journal articles** and a **minimum of 4 conference papers** in a dedicated References section at the end of your report (with correct APA citation style). In addition, this section needs to be at least 600 words, **excluding** the references.
- **Task 4.5 Clarification #1:** For this task, 8 points will be allocated to the comprehensiveness and accuracy of your literature review itself. The remaining 7 points will be allocated to the correctness & appropriateness of your references.
- **Task 4.5 Clarification #2:** Links to some websites do ***not*** count as a journal article or a conference paper reference!
  - **Example of a journal article & its reference:**
    - Yum, H., Lee, B., and Chae, M. (2012). From the wisdom of crowds to my own judgment in microfinance through online peer-to-peer lending platforms. *Electronic Commerce Research and Applications*, Vol 11, pp. 469–483.
  - **Example of a conference paper & its reference:**
    - Backstrom, L., Huttenlocher, D., Kleinberg, J., and Lan, X. (2006). Group formation in large social networks: Membership, growth, and evolution. *In 12th ACM International Conference on Knowledge Discovery and Data Mining,* pp. 44–54.
- **(Task 5) (10 points) Summary & Conclusions** of the **first phase** of your project: a comprehensive summary of Phase 1 and any insights you gained in this phase as they relate to your goals and objectives.

- **(Task 6) (REQUIRED, 3 points) References** For citing any book, article, paper, website, etc. in your report, you need to use the **APA style**.

RMIT Electronic Submission of Work for Assessment

When you submit work electronically at RMIT, you will be asked to agree to the **Assessment Declaration**.